

Salesforce Developer Bridge Program

Day 3 - Self Learning Track Report

Placement Management System

Student Details

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Project: Placement Management System

Day: Day 3

Technology: Salesforce Platform

Organization: Salesforce Developer Bridge Program

Day 3 Objective

The objective of Day 3 was to understand Salesforce Automation tools and their business applications. The focus was on learning Validation Rules, Record Triggered Flows, and Apex Triggers. Practical implementation was performed using Flow Builder to automate field updates, and interview preparation was completed by studying automation scenarios.

Task 1 - Understanding Validation Rules, Record Triggered Flow and Apex Trigger

Description

During this task, I studied the three major automation tools available in Salesforce.

Validation Rules were used to validate user input before saving a record.

Record Triggered Flow was studied as Salesforce's declarative automation tool which performs business automation without writing Apex code.

Apex Trigger was studied for implementing complex business logic whenever Flow cannot satisfy the business requirement.

I also learned Salesforce's recommended automation order:

Validation Rule

|

v

Flow

|

v

Apex Trigger

Learning Outcome

- Understood Validation Rules.
- Learned Record Triggered Flow.
- Learned Apex Trigger.
- Understood Salesforce Automation hierarchy.
- Learned when to choose each automation tool.

Screenshot

[Insert Screenshot Here – Validation Rule vs Flow vs Trigger]

Task 2 – Business Scenario Analysis

Description

Different business scenarios were analyzed to determine the correct Salesforce automation solution.

Scenario 1

Whenever a student submits a Job Application, send an email notification to the Placement Officer.

Solution: Record Triggered Flow

Scenario 2

Automatically populate Application Date.

Solution: Before Save Record Triggered Flow

Scenario 3

Prevent duplicate applications.

Solution: Validation Rule

Scenario 4

Reject applications when CGPA is below the required value.

Solution: Validation Rule

Scenario 5

Automatically create Offer Letter when Status becomes Selected.

Solution: After Save Record Triggered Flow

Learning Outcome

- Learned automation decision making.
- Improved business requirement analysis.
- Understood Salesforce best practices.
- Learned declarative automation.

Screenshot

[Insert Screenshot Here – Business Scenario]

Task 3 – Record Triggered Flow

Description

A Record Triggered Flow was created on the Job Application object. The Flow executes whenever a new Job Application record is created. The following configuration was completed.

- Object: Job Application
- Trigger: A record is created.
- Flow Type: Before Save Record Triggered Flow

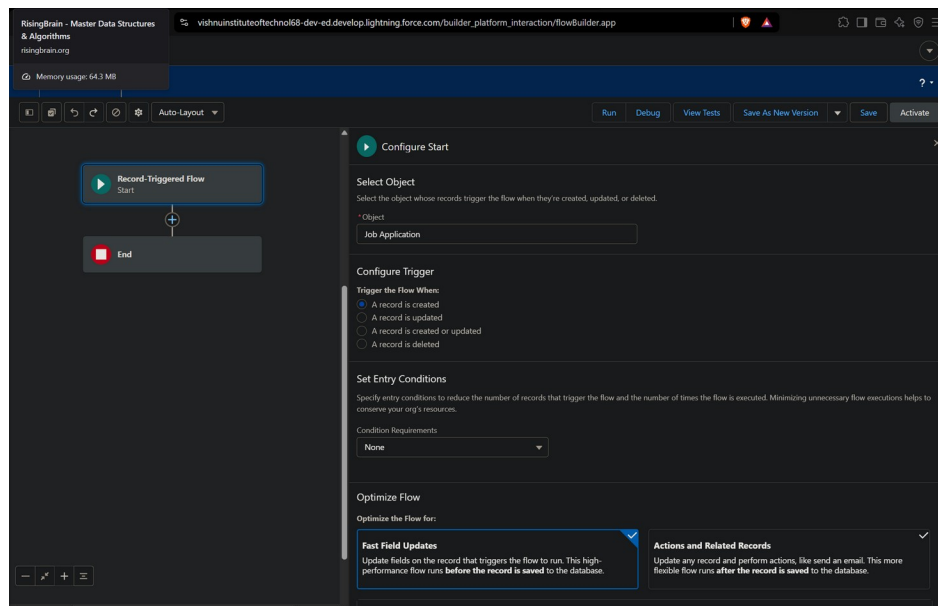


Figure 1: Flow Canvas & Start Element – Object: Job Application, Trigger: A Record Is Created (Before Save)

Assignment Element

An Assignment element was added. A custom field named Application Date was created. The Assignment element automatically populates the Application Date using today's date whenever a Job Application record is created.

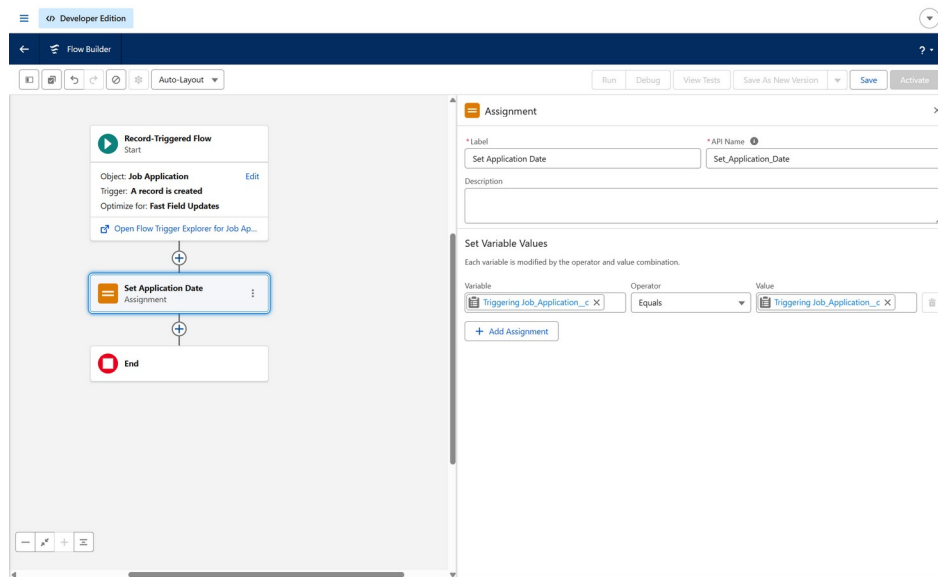


Figure 2: Assignment Element – Set Application Date (Initial Setup)

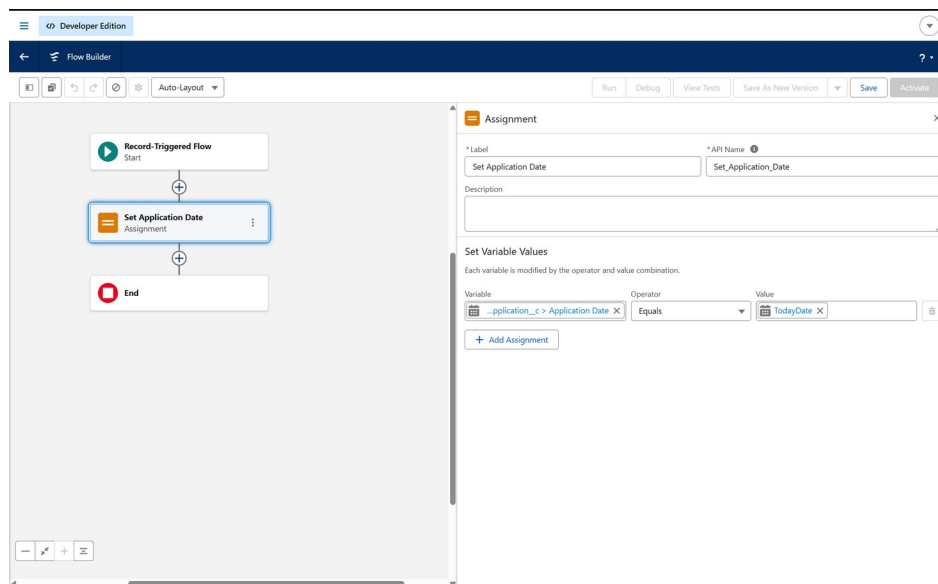


Figure 3: Assignment Element – Application Date Set to TodayDate

The Flow was successfully saved and activated.

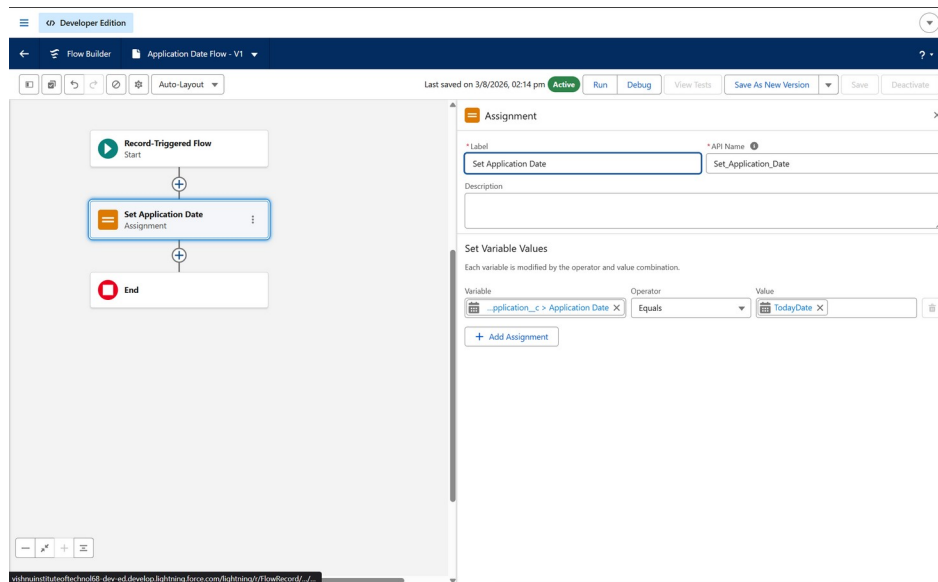


Figure 4: Application Date Flow – Saved and Activated

Learning Outcome

- Learned Flow Builder.
- Learned Before Save Flow.
- Learned Assignment Element.
- Learned automatic field updates.
- Successfully activated a Salesforce Flow.

Task 4 – Flow Implementation Challenges and Platform Limitations

Description

The Day 3 assignment required implementing an automated email notification using a Record Triggered Flow. The objective was to send an email to the Placement Officer whenever a new Job Application record was created.

During the implementation process, the Record Triggered Flow was successfully configured to automate the Application Date field. The Flow was created using Flow Builder, configured to execute before the record was saved, and activated successfully after validation.

While attempting to implement the email notification feature, it was observed that the current Salesforce Developer Edition org did not provide the required "Send Email" Flow Action or "Email Alert" action. Since these actions were unavailable in the current environment, the email notification functionality could not be configured.

Instead of abandoning the task, the available Flow functionality was completed successfully. The limitation was analyzed to understand the differences between Salesforce editions and the availability of standard Flow actions.

This exercise also provided practical knowledge about how Salesforce features can vary depending on the organization type, edition, enabled features, and user permissions. It emphasized the importance of verifying platform capabilities before designing automation solutions.

Learning Outcome

- Successfully created and activated a Record Triggered Flow.
- Learned how to troubleshoot missing Flow actions.
- Understood that Salesforce features may differ between Developer Edition and Production organizations.
- Gained knowledge about Flow Action availability and platform limitations.
- Improved problem-solving skills by identifying alternative approaches when required features were unavailable.
- Developed a better understanding of Salesforce automation architecture and deployment considerations.

Screenshot

[Insert Screenshot Here – Flow]

Task 5 – Salesforce Interview Preparation and Automation Concepts

Description

The final task of Day 3 focused on strengthening Salesforce interview knowledge by reviewing the concepts learned throughout the day's practical activities.

The interview preparation covered the purpose and working of Validation Rules, Record Triggered Flows, Before Save Flows, After Save Flows, and Apex Triggers. Special emphasis was placed on understanding when each automation tool should be used in real-world business scenarios.

Different automation scenarios were analyzed to determine the most appropriate Salesforce solution based on business requirements, scalability, maintainability, and platform best practices.

Topics Reviewed in Detail

- Validation Rules and their role in enforcing business constraints before records are saved.
- Record Triggered Flows and their ability to automate business processes without writing Apex code.
- Before Save Flows for efficiently updating fields before database operations.
- After Save Flows for creating related records, sending notifications, and performing post-save automation.
- Differences between Validation Rules, Flows, and Apex Triggers.
- Salesforce's recommended automation hierarchy, where declarative automation is preferred before implementing Apex.
- Best practices for designing scalable and maintainable automation solutions.

Common Interview Questions Practiced

- What is a Validation Rule?
- What is a Record Triggered Flow?
- Difference between Before Save Flow and After Save Flow.
- Difference between Flow and Apex Trigger.
- When should Apex Trigger be used instead of Flow?
- Why are Validation Rules important?
- Salesforce Automation Best Practices.
- Governor Limits and their impact on automation.
- Advantages of declarative automation.
- Real-world examples of Flow implementation.

Sample Interview Q&A

1. What is Validation Rule?

Validation Rule validates user input before saving a record.

2. What is Record Triggered Flow?

Record Triggered Flow automatically executes when a record is created, updated or deleted.

3. Difference between Before Save Flow and After Save Flow?

Before Save: updates the current record and runs faster. After Save: creates related records, sends emails, and calls Apex.

4. When should Flow be preferred over Trigger?

Whenever declarative automation can satisfy the requirement, Flow should be preferred.

5. When should Apex Trigger be used?

When complex business logic cannot be implemented using Flow.

Learning Outcome

- Strengthened understanding of Salesforce automation concepts.
- Improved confidence in answering Salesforce interview questions.
- Learned the recommended approach for selecting the appropriate automation tool.
- Understood Salesforce best practices for declarative automation.
- Improved knowledge of real-world business automation scenarios.
- Developed interview-ready explanations for Validation Rules, Record Triggered Flows, and Apex Triggers.
- Built a strong conceptual foundation for future Salesforce development and certification preparation.

Challenges Faced

- Application Date field was not available initially.
- Created the required custom Date field.

- Flow Email Action was unavailable in the current Salesforce Developer Edition org.
 - Successfully completed the available Flow implementation.
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Key Learnings

- Validation Rules improve data quality.
 - Record Triggered Flow automates business processes without code.
 - Apex Trigger handles complex business requirements.
 - Salesforce recommends using declarative automation before Apex.
 - Learned Flow Builder.
 - Learned Assignment Element.
 - Learned Flow Activation.
 - Improved interview preparation.
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Best Practices

- Prefer Validation Rules for data validation.
 - Prefer Flow before writing Apex.
 - Keep automation simple.
 - Follow Salesforce recommended automation hierarchy.
 - Build reusable automation.
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Summary

Day 3 focused on Salesforce Automation. I learned Validation Rules, Record Triggered Flows, and Apex Triggers. Business scenarios were analyzed to determine the appropriate automation tool. A Before Save Record Triggered Flow was successfully created on the Job Application object. The Flow automatically populates the Application Date field and was successfully activated.

Although the Flow Email Action was unavailable in the current Developer Edition org, the remaining practical activities were completed successfully.

Overall, Day 3 improved my understanding of Salesforce Automation, business process implementation, and interview preparation.